Technology Stack

Date	27 June 2026
Team ID	xxxxxx
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	*HTML5, CSS3, Bootstrap, JavaScript (Jinja2 templating)*	*Server-rendered templates integrate directly with Flask, keeping the app lightweight while Bootstrap ensures a responsive, mobile-friendly interface for viewing flood risk alerts and dashboards.*

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*Python (Flask)*	*Flask is lightweight and flexible, allowing rapid development of REST routes for authentication, chatbot integration, and flood prediction, while integrating seamlessly with Python-based ML libraries (scikit-learn/pandas) used for the prediction model.*
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*PostgreSQL*	*Provides strong relational structure needed for user accounts, OAuth/email verification records, and historical flood/weather data, along with reliable support for schema migrations as the app evolved.*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*Render*	*Offers straightforward, cost-effective hosting for Flask applications with managed PostgreSQL, automatic deploys from GitHub, and easy environment-variable configuration for API keys.*
5	**Version Control & CI/CD** *(e.g., GitHub, GitLab, Jenkins)*	*Git & GitHub*	*Enables branch-based collaboration and version tracking, with Render's GitHub integration allowing automatic redeployment on every push for continuous delivery.*
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*Google OAuth, Brevo (email API), Google Gemini API*	*Google OAuth enables secure one-click sign-in; Brevo powers transactional emails for signup and phone/email verification; the Gemini API drives the in-app chatbot widget that answers user queries about flood risk.*